

RAHUL KAKTIKAR

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OBJECTIVE

Electronics & Communication Engineering student specializing in **Embedded Systems** and **VLSI/Digital Design**. Hands-on experience with **ARM Cortex-M3**, **STM32** microcontroller programming, **Verilog HDL**, analog signal conditioning, and **PCB design (KiCad)**. Seeking an Embedded Systems / VLSI internship to apply and grow these skills in a product engineering environment.

TECHNICAL SKILLS

- Languages:** C (Embedded C), C++, Verilog HDL
- Microcontrollers & Architectures:** STM32F103 (ARM Cortex-M3), LPC2148 (ARM7TDMI), ESP32, ESP8266, Arduino Uno/Mega
- Embedded Peripherals:** GPIO, UART, SPI, I2C, Timers/PWM, ADC/DAC Interfacing
- VLSI / Digital Design:** Verilog HDL, FPGA Prototyping, Combinational & Sequential Logic Design
- Analog & PCB Design:** Op-Amp Signal Conditioning, Active Filter Design (Sallen-Key, Twin-T Notch), Wheatstone Bridge Sensing, PCB Schematic and Layout (KiCad)
- Tools & Software:** MATLAB/Simulink, KiCad, STM32CubeIDE, Git/GitHub, xilinx vivado, Keil µVision 5

ACADEMIC PROJECTS

Precision Temperature Controller | Analog & Embedded Signal Chain

- Designed an end-to-end analog signal-conditioning chain — NTC thermistor, Wheatstone bridge, and **LM358 op-amp** stages — for precision temperature sensing.
- Implemented a Twin-T 50 Hz notch filter and Sallen-Key low-pass filter ($f_c = 10.6$ Hz) for noise rejection, and interfaced a 12-bit **MCP3204 SPI ADC** with Arduino for digital acquisition.

Digital Logic Design Suite | VLSI / FPGA — Verilog HDL

- Designed and simulated combinational and sequential circuits in **Verilog HDL**, including multiplexers, flip-flops, adders, code converters, and sequence detectors.
- Built an FPGA-based Digital Dice Generator using pseudo-random logic implemented in Verilog.

8-bit R-2R DAC & Audio Frequency Analyzer | Mixed-Signal / Analog Design

- Designed an 8-bit R-2R resistor-ladder DAC for digital-to-analog conversion and validated output linearity.
- Built an audio frequency analyzer circuit using analog signal-processing techniques.

CERTIFICATIONS

- Simulink Onramp** — MathWorks Training Services (100% completion), 2026
- C Programming Basics** — SimpliLearn SkillUp, 2026

ACHIEVEMENTS & EXTRACURRICULAR

Participated in **“Buildverse”** Hackathon, KLE Technological University, BVB Campus, Hubli

Participated in **“Resonance”** Hackathon, Dr. MS Sheshgiri College of Engineering & Technology

EDUCATION

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| • KLE Technological University, Dr. M.S.Sheshgiri College of Engineering and Technology
<i>B.E. in Electronics and Communication Engineering</i> | Expected 2028
Belagavi, Karnataka |
| • Motichand Lengade Bhartesh Polytechnic
<i>Diploma in Electronics and Communication Engineering</i> | 2025
Karnataka |